



Real Estate, Unified.

The trust-first property platform for buyers, sellers, landlords, tenants, agents and investors — one account, one identity, every dashboard you need.

System Overview & 18-Week Roadmap · Generated 2026-06-30

1. Problems Vestra solves

Kenya's real-estate market loses billions every year to fraud, opaque pricing, paper leases that vanish, and chaotic rent collection. Vestra closes those gaps with one unified, trust-graded platform.

Buyers

- Forged title deeds & ghost sellers
- No transparent price-per-sqft data
- Paying cash to unverified parties
- No escrow culture — wire-and-pray

Sellers

- Cannot reach verified buyers directly
- No analytics on listing performance
- Commission opacity with brokers

Landlords

- Rent across cash / M-Pesa / bank — chaos
- Word-doc leases over WhatsApp
- Maintenance lost in phone calls
- No tax-ready rent roll

Tenants

- No proof of payment for visa/credit apps
- Maintenance complaints ignored
- Lease disputes — paper-only
- Heavy upfront move-in costs

Agents

- Leads on WhatsApp / Excel — no CRM
- Commission disputes with brokerages
- EARB license not visible to consumers

Investors (Chama)

- Group purchases tracked on chalkboards
- No transparent contribution ledger
- Decisions on phone calls — no voting

2. Differentiators

Trust layer. AI title-deed verification (OCR + Ministry of Lands cross-check), Trust Score 0–100 per property, EARB-verified Agent badges, multi-party escrow wallet.

Multi-role identity. One Vestra account, activate roles as life evolves: Member → Landlord → Seller → Investor → Tenant. Switchable from topbar / sidebar.

Native M-Pesa rails. STK Push for rent, deposits, escrow funding. Auto receipts (PDF) + tax-ready CSV. Paybill auto-reconciliation.

Chama Cooperative Engine. Shared cap table, M-Pesa-anchored contribution ledger, in-app voting, buy-out calculator.

Lease-as-a-Service. Template marketplace + e-sign + hash-anchored timestamps + dispute timeline.

Maintenance Marketplace. Vetted contractor pool, photo-required updates, SLA timer, micro-escrow.

Investment Intelligence. Suburb price index, rental yield calculator, comparable sales w/ confidence intervals.

3. Feature Roadmap by Role

3.1 Member (base)

- **SHIPPED** Discoverable sidebar — Market, Verify, Insights, Investments, Escrow
- **PLANNED** Saved searches with email/SMS alerts
- **PLANNED** Personalized recommendations
- **FUTURE** Trust circle — invite past transaction partners to vouch

3.2 Seller

- **SHIPPED** Multi-role activation + listing CRUD (backend wired)
- **PLANNED** Listing analytics — views over time, inquiry funnel
- **PLANNED** Featured listing boost (paid)
- **PLANNED** Buyer Q&A thread on each listing
- **FUTURE** Photographer marketplace

3.3 Landlord

- **SHIPPED** Unit CRUD + tenant invite flow (e2e: invite → accept → role activates)
- **PLANNED** Rent roll table w/ month picker
- **PLANNED** Auto rent-reminder SMS
- **PLANNED** Vacancy → one-click main-portal listing
- **PLANNED** Tenant scoring across the platform

- **PLANNED** KRA-ready tax export
- **PLANNED** Maintenance ticket flow + contractor pool
- **FUTURE** Insurance broker tie-in

3.4 Tenant

- **SHIPPED** Invite acceptance + auto role activation
- **PLANNED** Rent payment via M-Pesa STK
- **PLANNED** Receipt vault — PDF + bulk download
- **PLANNED** Maintenance request with photo upload
- **PLANNED** Lease viewer + e-sign
- **PLANNED** Move-out checklist + deposit return tracker
- **FUTURE** Credit-building report (rent on-time history → bank-grade letter)

3.5 Agent

- **SHIPPED** Profile + dashboard shell + backend wired
- **PLANNED** Lead pipeline (kanban: new → contacted → viewing → offer → closed)
- **PLANNED** Commission tracker w/ auto-pull from closed escrows
- **PLANNED** Subscription tier enforcement (see §4)
- **PLANNED** Featured agent placements
- **FUTURE** Co-broker split tracking, private deal rooms

3.6 Investor (Chama)

- **PLANNED** Group creation + member invites
- **PLANNED** Cap table + contribution ledger
- **PLANNED** Vote engine (majority / supermajority / unanimous)
- **PLANNED** Dividend distribution from rental income

3.7 Admin

- **SHIPPED** User table, verification queue, fraud reports shells
- **PLANNED** AI-flagged listing review queue
- **PLANNED** Disputed-transaction arbitration UI
- **PLANNED** Revenue dashboard

4. Subscriptions & Monetization

Strategy. Layered revenue: subscriptions (recurring) + take rate (transactional) + B2B referrals (commission). Pricing benchmarked to Kenyan SaaS willingness-to-pay.

4.1 Agent Tiers

Tier	KES / mo	Listings	Featured	Lead alerts	Co-broker	Badge
Free	0	5	0	Email only	—	Bronze
Starter	1,500	20	2/mo	Email + SMS	—	Silver
Pro	4,500	100	10/mo	Real-time + WhatsApp	Yes	Gold
Elite	12,000	Unlimited	Unlimited	Priority + CS manager	Yes + exports	Platinum

4.2 Landlord Tiers

Tier	KES / mo	Units	Auto-reminders	Maintenance	Tax export
Free	0	3	Manual	Basic	—
Plus	800	15	Yes	Yes	CSV
Pro	2,500	50	Yes + escalation	+ contractor mkt	CSV + PDF
Estate	8,000	Unlimited	+ tenant scoring	Dedicated coord	Accountant-ready

4.3 Take-rate revenue (deal-based)

Item	Rate
Escrow fee	0.5% of transaction, cap KES 100k
Verification report	KES 2,500 / property
Featured listing boost	KES 1,000–5,000 / week
Lease e-sign with stamp	KES 500 / document
Tenant credit-report letter	KES 200 / letter
Mortgage referral	0.25–0.5% of loan value (bank pays)
Insurance referral	5–10% commission (broker pays)

4.4 Buyer micro-payments

Tier	Price	What you get
Free	—	Browse + save 20 properties
Verified Buyer	KES 500 one-off	Unlimited saves, priority inquiry, identity-verified shield
Pro Buyer	KES 2,000 / mo	AI valuation, mortgage pre-qual, suburb deep-dives, ROI calculator

4.5 Chama tiers

Tier	KES / mo	Members	Properties	Voting	Audit log
Starter	0	10	1	Basic	30 days
Group	1,500	50	5	+ supermajority	1 year
Cooperative	5,000	Unlimited	Unlimited	+ custom rules	Unlimited

4.6 Year-1 revenue model

Assumptions — Kenya, year-1 traction: 5,000 active agents · 15,000 landlords managing ~80k units · 100k buyers · 1,000 chamas.

Source	MRR (KES)
Agent subscriptions	~4.2M
Landlord subscriptions	~5.6M
Chama subscriptions	~1.0M
Buyer Pro	~4.0M
Escrow take rate	~5.0M
Verification + e-sign + alerts	~1.5M
Total MRR	~KES 21M (~USD 145k)
ARR	~KES 250M (~USD 1.74M)

5. UI Modernization

- **SHIPPED** Dark mode (Tailwind v4 with @custom-variant)
- **SHIPPED** Code splitting (recharts/maps/framer/icons vendors)
- **SHIPPED** Lazy heavy components + idle-callback mount
- **SHIPPED** Persistent role-scope breadcrumb on every dashboard page
- **PLANNED** Motion library — micro-interactions on every CTA
- **PLANNED** Skeletons on every data-fetch surface
- **PLANNED** Empty states designed per feature
- **PLANNED** Global command palette (Cmd-K) — search across listings, units, leases
- **PLANNED** Smart inbox: messages + notifications + invites merged
- **PLANNED** "What's next" widget on home
- **PLANNED** WCAG 2.2 AA pass + keyboard nav + screen reader testing
- **FUTURE** Image CDN + responsive srcset + service worker offline shell

6. Mobile Path — Play & Apple Stores

Recommended architecture: Capacitor. Wraps the existing Vite/React PWA into native iOS + Android binaries. One codebase, native APIs available, ~2 weeks engineering for the wrap itself, then 6–8 weeks to submission and approval.

6.1 Setup commands

```
# 1. Install Capacitor
cd vestra_properties
npm install @capacitor/core @capacitor/cli @capacitor/ios @capacitor/android
npx cap init "Vestra" "com.vestra.app" --web-dir=dist

# 2. Build web bundle
npm run build

# 3. Add native platforms
npx cap add ios
npx cap add android
npx cap sync

# 4. Open in native IDEs
npx cap open ios      # Xcode
npx cap open android  # Android Studio
```

6.2 Native plugins to add

- `@capacitor/push-notifications` — rent reminders, invite alerts
- `@capacitor/camera` — property/maintenance photos
- `@capacitor/geolocation` — map filter, suburb detection
- `@capacitor/share` — listing share to WhatsApp
- `@capacitor-community/biometric-auth` — login + STK confirm
- `@capacitor/local-notifications` — offline rent due reminders
- M-Pesa Daraja SDK (community plugin or REST direct)

6.3 Store submission checklist

Apple App Store

- Apple Developer Account (\$99/yr)
- App Store Connect listing — name, subtitle, keywords, screenshots @ 6.5" / 5.5"
- Privacy policy URL
- App Privacy questionnaire (PII, location, financial data)
- TestFlight beta — 25+ Kenya testers
- Review: 24–72 hr typical

Google Play

- Play Console (\$25 one-off)
- Short + full description, feature graphic, 8 screenshots
- Content rating questionnaire
- Data safety declaration
- Closed testing — 20+ testers × 14 days
- Closed → open → production release

6.4 Compliance landmines

- **Financial services flag** — Apple may demand banking license disclosure if M-Pesa rent is in-app. Mitigation: position as "third-party payment integration," surface clear T&Cs.
- **Real-estate brokerage license** — visible reference to EARB compliance required.
- **KYC privacy** — disclose ID collection in Privacy Policy.

6.5 Timeline

Weeks	Milestone
1–2	Capacitor wrap, native plugin integration, dev builds
3–4	Internal testing (Vestra team)
5	TestFlight + Play closed testing
6–7	Store listings, screenshots, privacy policy
8	Submit. Apple ~3 days. Play ~14 days.

Realistic public launch: ~10 weeks from start of mobile work.

7. Shipped vs Outstanding

Shipped this build cycle

- **SHIPPED** Multi-role architecture (User.roles[] + activeRole) backed by Postgres
- **SHIPPED** Role activation modal wizard for self-serve roles
- **SHIPPED** Roles Hub at /dashboard/roles

- **SHIPPED** Role switcher in topbar + sidebar
- **SHIPPED** Tenant invite flow end-to-end (landlord → tenant → auto-role)
- **SHIPPED** Persistent dashboard scope bar
- **SHIPPED** Sidebar discoverability — Investments, Escrow, Insights, Roles visible
- **SHIPPED** Real Node + Express + Prisma backend (~30 endpoints)
- **SHIPPED** Postgres schema, 14 models, seeded demo data
- **SHIPPED** Frontend fully wired to backend
- **SHIPPED** Tailwind v4 dark mode fix
- **SHIPPED** Vite perf tuning (vendor chunks, optimizeDeps, idle-callback)
- **SHIPPED** TS clean (exit 0 both ends)

Outstanding (next 4.5 months)

- **PLANNED** Subscription tier enforcement (backend + paywall UI)
- **PLANNED** M-Pesa Daraja integration (sandbox → live)
- **PLANNED** AI verification pipeline (OCR + KENHA cross-check)
- **PLANNED** Chama cooperative module
- **PLANNED** Maintenance contractor marketplace
- **PLANNED** Lease e-sign service
- **PLANNED** Notification preferences (SMS, email, in-app)
- **PLANNED** Saved searches + alerts
- **PLANNED** Capacitor mobile wrap + native plugins
- **PLANNED** Store submissions
- **PLANNED** WCAG AA accessibility audit
- **PLANNED** Service worker / offline shell
- **PLANNED** Production hardening — Sentry, audit log, per-route rate limits

8. Phased Build Plan

Phase	Scope	Weeks
0	Foundations + multi-role + tenant invite (DONE)	—
1	Subscription tiers + paywall enforcement	2
2	M-Pesa STK live + escrow flows	2
3	Chama cooperative module	3
4	Maintenance marketplace + lease e-sign	3
5	AI verification (MVP — OCR + manual review)	2
6	Capacitor wrap + native plugins	2
7	Store submission + closed testing	4
8	Public launch + paid acquisition	ongoing
Total focused engineering to "world-class + on stores"		~18 weeks (~4.5 months)

9. KPIs & Risks

KPIs to track

- **Activation** — % of signups who activate ≥1 non-buyer role within 7 days
- **Subscription conversion** — free → paid by role
- **NRR** — keep > 100% via tier upgrades + take rate
- **Transaction GMV** — escrow + rent processed / month
- **Trust Score avg** — > 85 across active listings
- **Maintenance SLA** — % tickets closed within stated window
- **DAU/MAU** — > 25%
- **Mobile share** — > 70% within 6 months of native launch

Risks & mitigations

Risk	Mitigation
Apple flagging M-Pesa as financial product	Partner with licensed PSP, route via their SDK
Real-estate license requirements vary	Auto-route listings via EARB-verified agents; tag DIY clearly
Title-deed verification false negatives	Human-in-the-loop review for trust score < 80
Deposit return disputes	Micro-escrow w/ photo-audit auto-release
Competition (BuyRentKenya, Hass, Property24)	Differentiate on trust + role unification, not just listings
Chama legal grey area	Engage cooperative-savings lawyer (SACCO Act compliance)

10. What to do next — step by step

1. **Run what's shipped.** Backend on :4000 + frontend on :5173 already wired. Login as `landlord@vestra.com` / `password`, add a unit, invite `buyer@vestra.com`, accept the invitation. Full e2e works.
2. **Pick Phase 1 — subscriptions.** Add `Subscription` Prisma model, paywall middleware, UI tier picker in agent + landlord workspaces, M-Pesa subscription billing via Daraja sandbox.
3. **Then M-Pesa rails (Phase 2).** Daraja sandbox keys, STK push for rent, escrow funding.
4. **Phase 3 — Chama.** Group + cap-table + voting models.
5. **Phase 4 — Maintenance + lease e-sign.**
6. **Phase 5 — AI verification.** OCR via Tesseract for MVP, upgrade to dedicated model later.
7. **Phase 6–7 — Mobile.** Capacitor wrap, plugins, store assets, submissions.
8. **Phase 8 — launch + marketing.** Paid Meta + Google Ads, agent acquisition push, referral program.